

Process Design Basis

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1 SITE CONDITIONS

The site is located in the Republic of Kazakhstan by the northern coast of the Caspian Sea within 50 km of the city of Atyrau. The territory of the Atyrau region represents a low flat country or a slightly elevated flat country, located in a semi-desert and desert zone.

The Power Plant shall be able to supply electric power and HP steam to its design capacity, in the summer and winter conditions specified hereafter, as well as in any intermediate climatic condition.

		Winter conditions	Average conditions	Summer conditions
Ambient temperature	°C	-30	8.6 (note 1)	40
Atmospheric pressure	mbar	1023.5	1013	992
Relative humidity	%	82	62	46

Note 1: monthly average on 1986-2007 period

2 FLUID CONDITIONS

The chemical characteristics of the fluids flowing in the thermal cycle of Combined Cycle Power Plant are based on VGB R 450 Le "Guidelined for Feed Water, Boiler Water and Steam Quality for Power Plants/Industrial Plants.

The Normal Level (N) of the following tables define respectively the chemical characteristics of feedwater, boiler water and steam.

Table 6.5: Drum Boilers with a copper free condensate / feed water system

FEED WATER and Attemperator Spray Water						
Boiler Type	Drum; copper free condensate / feed water system					
Feed Water Treatment			AVT (Alkaline)		OT	
pH ⁽¹⁾		N	9.2 – 9.5		8.8 – 9.2	
		AL 1	9.2	9.8	8.6	9.8
		AL 2	9.0	10.0	8.4	10.0
		AL 3	8.6	10.4	8.2	10.4
Conductivity (only valid for ammonia dosing)	µS/cm	N	4.3 – 8.6		1.7 – 4.3	
		AL 1	4.3	18	1.1	18
		AL 2	2.8	28	0.7	28
		AL 3	1.1	68	0.4	68
Acid Conductivity	µS/cm	N	< 0.10		< 0.10	
		AL 1	0.20 ⁽²⁾		0.15	
		AL 2	0.50		0.20	
		AL 3	1		0.50	
Oxygen (O ₂)	µg/kg	N	5 – 20		> 30	
		AL 1	2 - 100		100	
		AL 2	> 250		250	
		AL 3	-		1000	
Silica (SiO ₂)	µg/kg	N	< 5		< 5	
		AL 1	20		20	
		AL 2	50		50	
		AL 3	–		–	
Iron (Fe), total	µg/kg	N	< 10		< 10	
		AL 1	20		20	
		AL 2	30		30	
		AL 3	–		–	
Organics	see chapter 7.10					

N: Normal Level
AL 1 (2, 3): Action Level

⁽¹⁾ Use of oxygen scavengers see chapter 7.3

⁽²⁾ A higher value up to AL 2 may be acceptable if the increase of acid conductivity can be attributed to carbon dioxide

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Table 6.10: Steam

STEAM for steam turbines				
Acid Conductivity ⁽¹⁾	μS/cm	N	0.10	
		AL 1	0.20	
		AL 2	0.50	
		AL 3	1	
Silica (SiO ₂)	μg/kg	N	< 5	
		AL 1	20	
		AL 2	–	
		AL 3	–	
Sodium (Na)	μg/kg		Boiler Water Treatment AVT or Caustic	Boiler Water Treatment Phosphate
		N	< 2	< 5
		AL 1	5	10
		AL 2	10	20
		AL 3	20	40
Iron (Fe), total	μg/kg	N	< 5	
		AL 1	20	
		AL 2	–	
		AL 3	–	
Copper (Cu), total	μg/kg	N	< 1	
		AL 1	3	
		AL 2	–	
		AL 3	–	

N: Normal Level

AL 1 (2, 3): Action Level

⁽¹⁾ Higher action values may be defined if the increase of acid conductivity can be attributed to carbon dioxide from air leakage and organic decomposition products can be excluded.

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Table 6.8: Drum Boilers with Phosphate Treatment

BOILER WATER								
Boiler Type		Drum						
Boiler Water Treatment			Phosphate Treatment					
Steam Pressure (MPa)			< 4		4 – 10		> 10	
pH		N	9.8 – 10.2		9.5 – 10.0		9.4 – 9.6	
		AL 1	9.5	10.5	9.4	10.2	9.3	9.7
		AL 2	9.0	10.7	9.0	10.3	9.0	9.9
		AL 3	8.5	11.0	8.5	10.5	8.5	10.2
Conductivity	μS/cm	N	25 – 100		15 – 50		10 – 30	
		AL 1	> 100		> 50		> 30	
		AL 2	250		100		50	
		AL 3	500		200		100	
Phosphate (PO ₄)	mg/kg	N	< 15		< 6		< 3	
Silica (SiO ₂)	mg/kg	N	0.25 x AL 1					
		AL 1	according to figure 4 respectively 5					
		AL 2	2 x AL 1					
		AL 3	–					
Organics	see chapter 7.10							

N: Normal Level
 AL 1 (2, 3): Action Level

The demineralized water coming from Water Treatment Plant must have an acid conductivity equal to 0.1 μS/cm

3 LINE SIZING CRITERIA

3.1 Liquid Lines

The piping containing liquid fluids shall be sized according to the guidelines of the following table:

	Velocity (notes 3,4) [m/s]	Pressure losses [bar/100m]
Pump suction lines (notes 1) - bubble point liquids - subcooled liquids - DN<8" - DN>8"	1.8 2.4 3.7	0.11 0.45 0.45
Pump discharge lines - carbon steel - stainless steel/alloy steel	6.1 7.6	0.91 0.91
Liquid transfer lines (note 2) Cooling water lines - main header - branches	3.7 3.7 3.7	0.34 0.15 0.30
Steam condensate lines		0.61

Notes

(1) Pump suction line diameters should normally not be more than two (2) standard line sizes larger than the pump suction nozzle.

(2) Or as required by system pressure balance.

(3) If the liquid velocity is too high, swaged up meter runs may be required. Velocities for lines containing orifice plates should be limited to the following values:

- for 2" and up, velocity < 3.4 m/s

- for 14" and up, velocity < 3.1 m/s.

(4) For reciprocating pump, acceleration head loss has to be taken into account in the pump suction circuit design (the installation of a damper may be required to avoid acceleration loss. To be studied with pump vendor).

3.2 Gaseous lines

The piping containing gaseous fluids shall be sized according to the guidelines of the following table:

	Velocity (note 1) [m/s]	Pressure losses [bar/100m]
Hydrocarbon lines:		
• vacuum (< 0.07 bara)	$122/p^{0.5}$	0.014
• vacuum (= 0.48 bara)	$122/p^{0.5}$	0.034
• 0.0 – 3.45 barg	$122/p^{0.5}$	0.11
• 3.45 – 10.3 barg	$122/p^{0.5}$	0.34
• 10.3 barg – 34.5 barg	$122/p^{0.5}$	0.68
• > 34.5 barg	$122/p^{0.5}$	1.13
Steam lines:		
• vacuum (< 0.07 bara)	-	0.014
• vacuum (= 0.48 bara)	-	0.034
• 0.0 – 3.45 barg	-	0.11
• 3.45 – 10.3 barg	-	0.34
• 10.3 barg – 34.5 barg	-	0.68
• > 34.5 barg	-	1.13

Notes

(1) vapour density (p) measured in kg/m³

4 EQUIPMENT SIZING CRITERIA

4.1 Deaerator

The deaerator shall be sized to take into account the maximum needs of Combined Cycle Power Plant (CCPP) and the worst case of IPC start up.

- high pressure steam flow rate during IPC start-up (circa 260 t/h)
- high pressure steam flow rate to feed CCPP steam turbine

The deaerator is fed by steam coming from different sources:

- low pressure steam coming from HRSGs continuous blowdown drums
- low pressure steam coming from LP auxiliary steam boiler

The deaerating pressure shall be set equal to 1 barg in order to be able to maximize the steam recovery from HRSGs continuous blowdown drums and to feed the deaerator with low pressure steam coming from auxiliary steam boiler.

The hold-up of the deaerator shall be fixed equal to 20 minutes of the maximum feedwater required from CCPP.

4.2 Main Pumps

Condensate Extraction Pumps

The condensate extraction pumps capacity shall be based on HRSGs working in fresh air mode during summer conditions.

The condensate extraction pumps shall be checked in order to allow them to manage the condensate flow in case of steam turbine bypass operation in every site condition.

The head of condensate extraction pumps shall be based on the pressure losses of the condensate through all the circuit that starts from the hotwell of the air cooled condenser and ends with the deaerator.

Boiler Feedwater Pumps

The boiler feedwater pumps capacity shall be based on the following conditions:

- HRSGs working in post firing mode during winter conditions
- maximum high pressure steam production to be exported to SEZ NIPT users (43 bar(g)) .

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A capacity margin equal to 10% has been taken into account in boiler feedwater pumps sizing.

The head of boiler feedwater pumps shall be based on the pressure losses of HRSG heating surfaces and on the maximum pressure at steam turbine inlet.

A margin equal to 3% on boiler feedwater pumps discharge shall be considered to take into account losses due to wear.